Calculating paper results

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Overview

This is just a quick note illustrating results from the "initial" simulations:

```
scenarios = expand.grid(
  K = c(10),
  J.bar = c(20,50),
  p.bar = c(0.5),
  ATE.district = 0.2,
  sd.district = 0.1,
  ICC.district = 0.2,
  variable.J = c(T,F),
  variable.p = c(T,F),

  # cluster-level params
  n.bar = c(50),
  sd.school = c(0.1),
  ICC.school = 0.2,
  variable.n = c(T,F),

  # size and correlation params
  size.impact.cor.district = c(0,0.75),
  size.impact.cor.school = c(0,1),
  proptx.impact.cor.district = c(0,0.75))
```

Key for estimand names:

- "stu": student-weighted
- "sch": school-weighted
- "dstu": district-weighted student-weighted
- "dsch": district-weighted school-weighted

Absolute bias

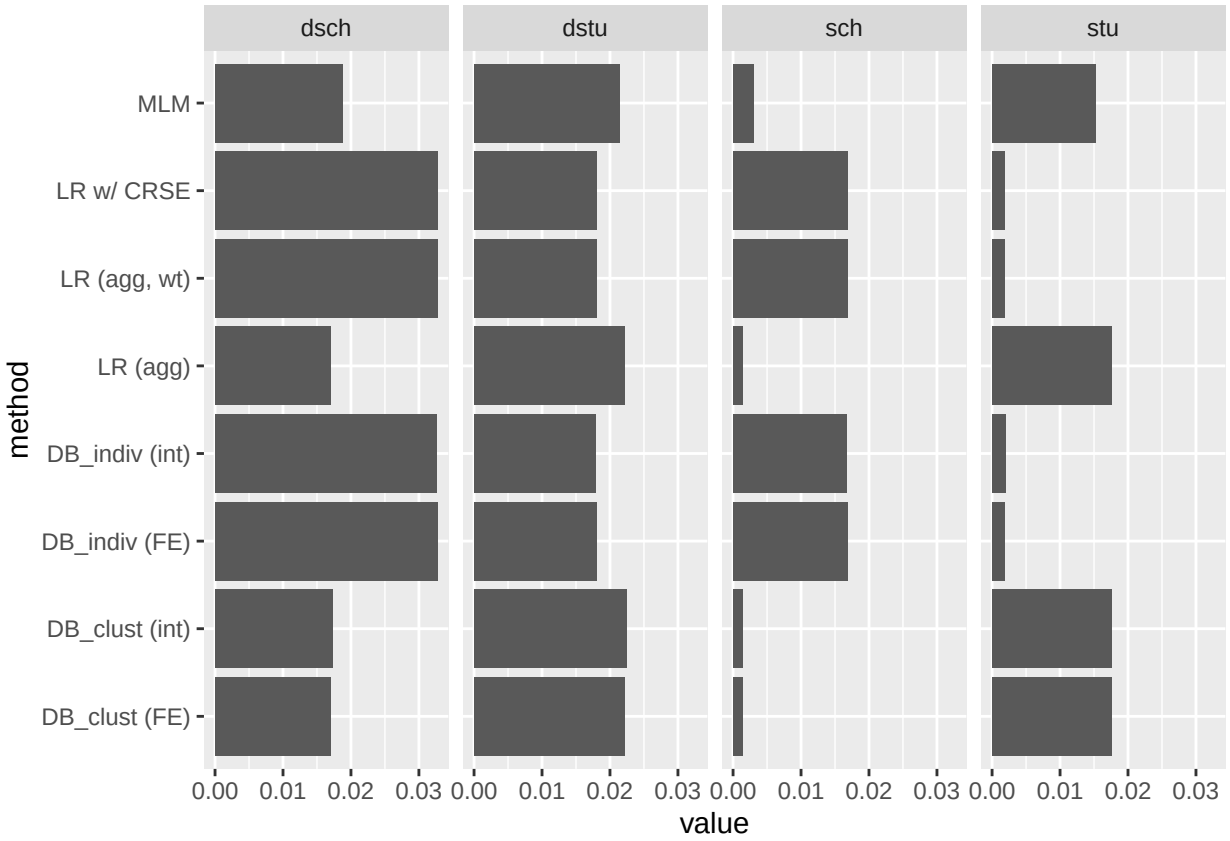
There are three groups of methods:

1. Student-weighted: LR w/ CRSE, LR (agg, wt), DB_indiv (int), DB_indiv (FE). These methods have less bias for the student-weighted estimands.
2. School-weighted: LR (agg), DB_clust (int), DB_clust (FE). These methods have less bias for the school-weighted estimands.
3. MLM: about 90% of the way down the spectrum from student-weighted to school-weighted (i.e., slightly different from school-weighted).

The differences between the student and school-weighted methods are proportionally smaller for the district-weighted versions of those estimands.

For clarity I've removed the bias for superpopulation estimands here, since true finite-sample and superpopulation estimand averages are the same.

```
sruns %>%
  group_by(method) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  filter(finite) %>%
  ggplot(aes(x=method, y=value)) +
  geom_col() +
  facet_grid( ~ estimand) +
  coord_flip()
```



Old tables:

```
bias.table
```

```
## # A tibble: 5 x 5
##   method         mean.bias.stu rng.bias.stu mean.bias.sch rng.bias.sch
##   <chr>                  <dbl> <chr>                <dbl> <chr>
## 1 DB_clust (int)     -0.0164   (-0.05-0.00)    0.00000996 (-0.00-0.00)
## 2 DB_indiv (int)     -0.000557 (-0.00-0.00)    0.0159     (-0.00-0.05)
## 3 LR (agg)           -0.0167   (-0.05-0.00)   -0.000227   (-0.00-0.00)
## 4 LR (agg, wt)       -0.000441 (-0.00-0.00)    0.0160     (-0.00-0.05)
## 5 MLM                -0.0144   (-0.04-0.00)    0.00208    (-0.00-0.01)
```

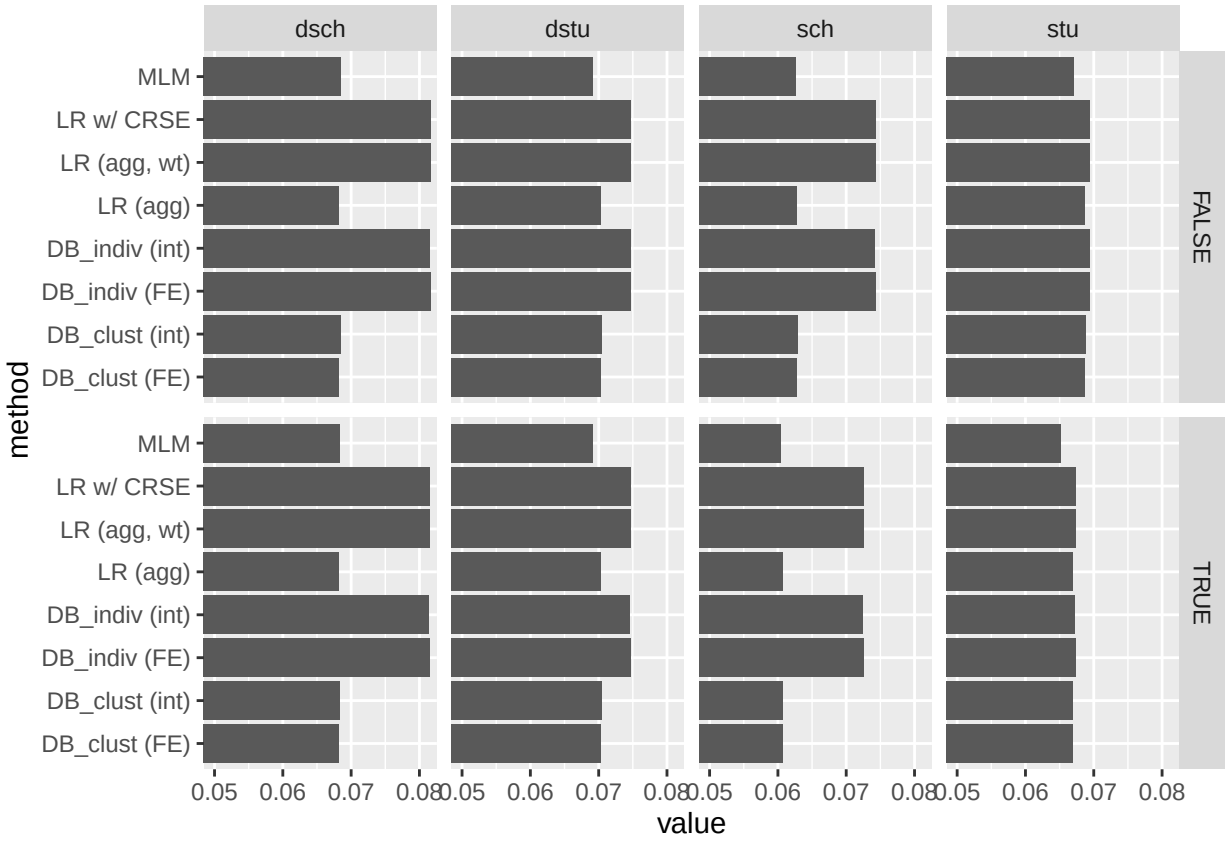
```
bias.table.pct
```

```
## # A tibble: 5 x 5
##   method         mean.bias.stu rng.bias.stu mean.bias.sch rng.bias.sch
##   <chr>                  <dbl> <chr>                <dbl> <chr>
## 1 DB_clust (int)        13.4   (0-55)              0.0847 (0-0)
## 2 DB_indiv (int)         0.131 (0-0)              10.6    (0-45)
## 3 LR (agg)              13.5   (0-55)              0.0859 (0-0)
## 4 LR (agg, wt)           0.118 (0-0)              10.7    (0-46)
## 5 MLM                   11.5   (0-48)              0.515  (0-2)
```

RMSE

Similar to bias results, but MLM is just uniformly slightly better than everything else, for every estimand.

```
sruns %>%
  group_by(method) %>%
  summarize(across(starts_with("RMSE"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("RMSE")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  ggplot(aes(x=method, y=value)) +
  geom_col() +
  facet_grid(finite ~ estimand) +
  coord_flip(ylim = c(0.05, 0.081))
```



Old tables:

```
RMSE.table
```

```
## # A tibble: 5 x 17
##   method    mn.f.stu rng.f.stu mn.s.stu rng.s.stu mn.f.dstu rng.f.dstu mn.s.dstu
##   <chr>        <dbl> <chr>        <dbl> <chr>         <dbl> <chr>          <dbl>
## 1 DB_clust~    101.  (84-131)     100.  (84-130)       96.0 (72-125)        95.9
## 2 DB_indiv~     99.9 (99-101)      99.9 (99-101)       99.9 (99-101)        99.9
## 3 LR (agg)     101.  (84-132)     100.  (84-131)       95.8 (72-126)        95.6
## 4 LR (agg,~    100   <NA>         100   <NA>          100   <NA>           100
## 5 MLM           97.9 (84-121)      97.6 (83-120)       94.0 (72-116)        93.9
## # i 9 more variables: rng.s.dstu <chr>, mn.f.sch <dbl>, rng.f.sch <chr>,
```

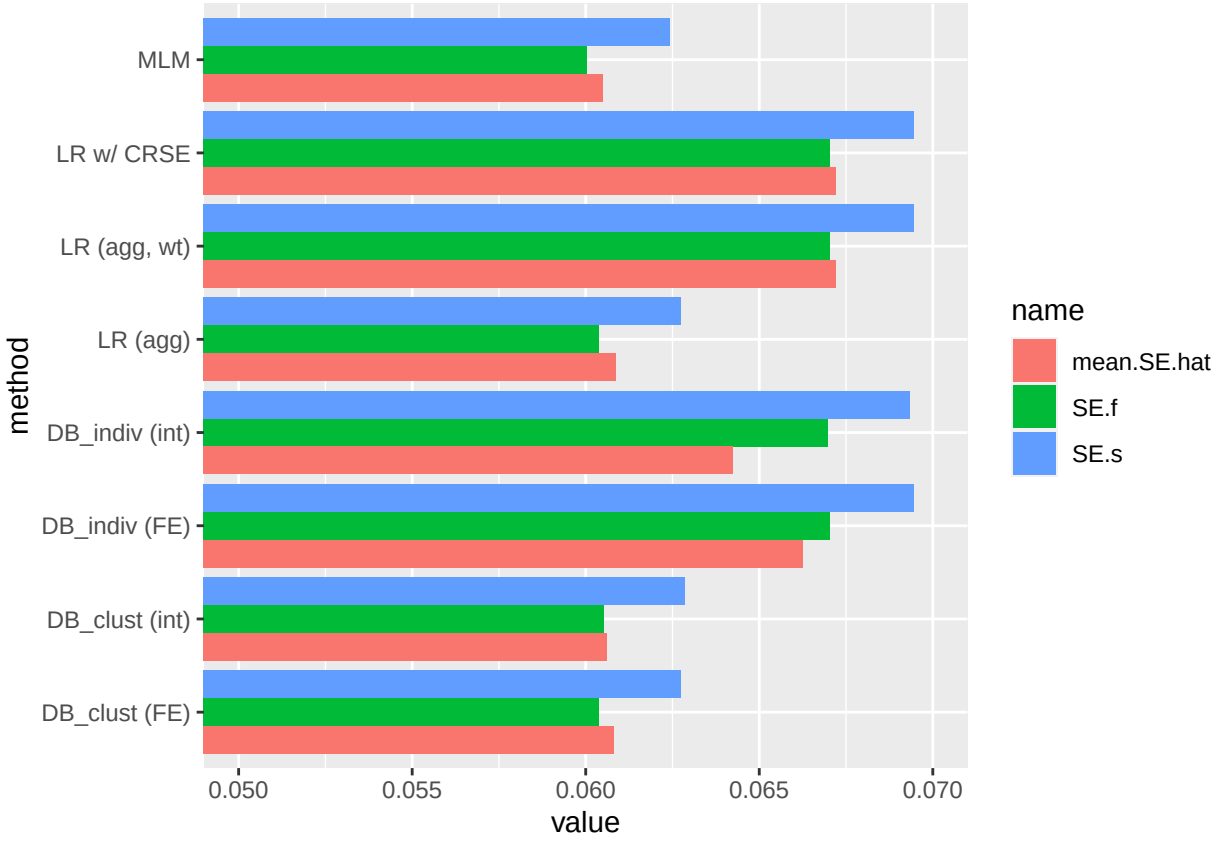
```
## #   mn.s.sch <dbl>, rng.s.sch <chr>, mn.f.dsch <dbl>, rng.f.dsch <chr>,
## #   mn.s.dsch <dbl>, rng.s.dsch <chr>
```

SE

The student-weighted methods generally have greater estimated and true standard errors.

In general, methods' estimated standard errors tend to be similar to their finite-sample standard errors. As expected, methods' true finite-sample standard errors are less than their true superpopulation standard errors.

```
sruns %>%
  group_by(method) %>%
  summarize(across(starts_with("SE."), ~mean(.)),
            mean.SE.hat = mean(mean.SE.hat)) %>%
  pivot_longer(contains("SE")) %>%
  ggplot(aes(x=method, y=value, group=name, fill=name)) +
  geom_col(position="dodge") +
  coord_flip(ylim=c(0.05, 0.07))
```



Note: unlike the results table below, the plot above is not normalized, so different simulation scenarios will have different true SEs.

Old tables:

```
SE.table
```

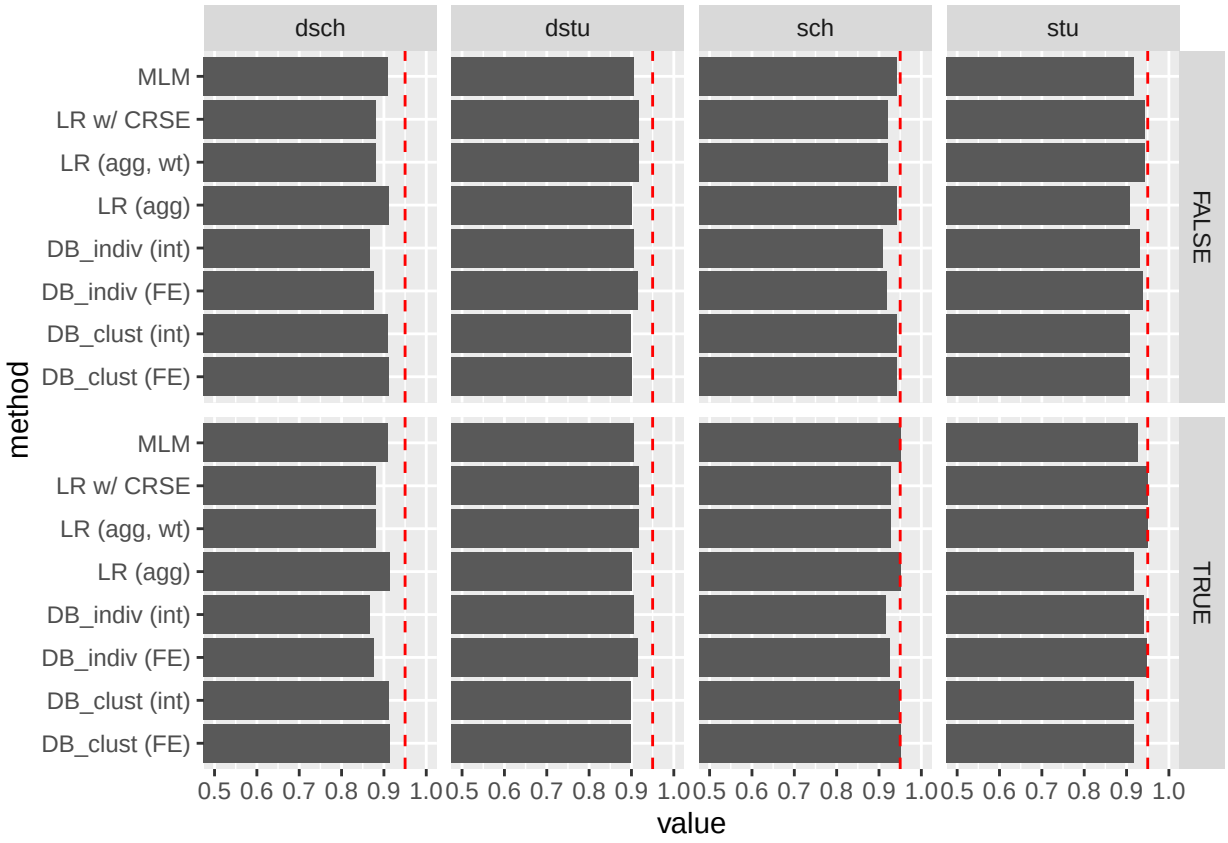
```
## # A tibble: 5 x 7
##   method          finite range.f  super range.s  inflation range.i
##   <chr>            <dbl> <chr>    <dbl> <chr>        <dbl> <chr>
## 1 DB_clust (int)   90.8 (84-101)  91.0 (83-101)      104. (99-115)
## 2 DB_indiv (int)  100.  (99-101)  99.9 (99-101)      104. (100-112)
```

```
## 3 LR (agg)         90.6 (83-100)  90.9 (83-100)      104. (99-115)
## 4 LR (agg, wt)    100   <NA>     100   <NA>          104. (100-112)
## 5 MLM              90.1 (83-100)  90.4 (83-100)      104. (99-115)
```

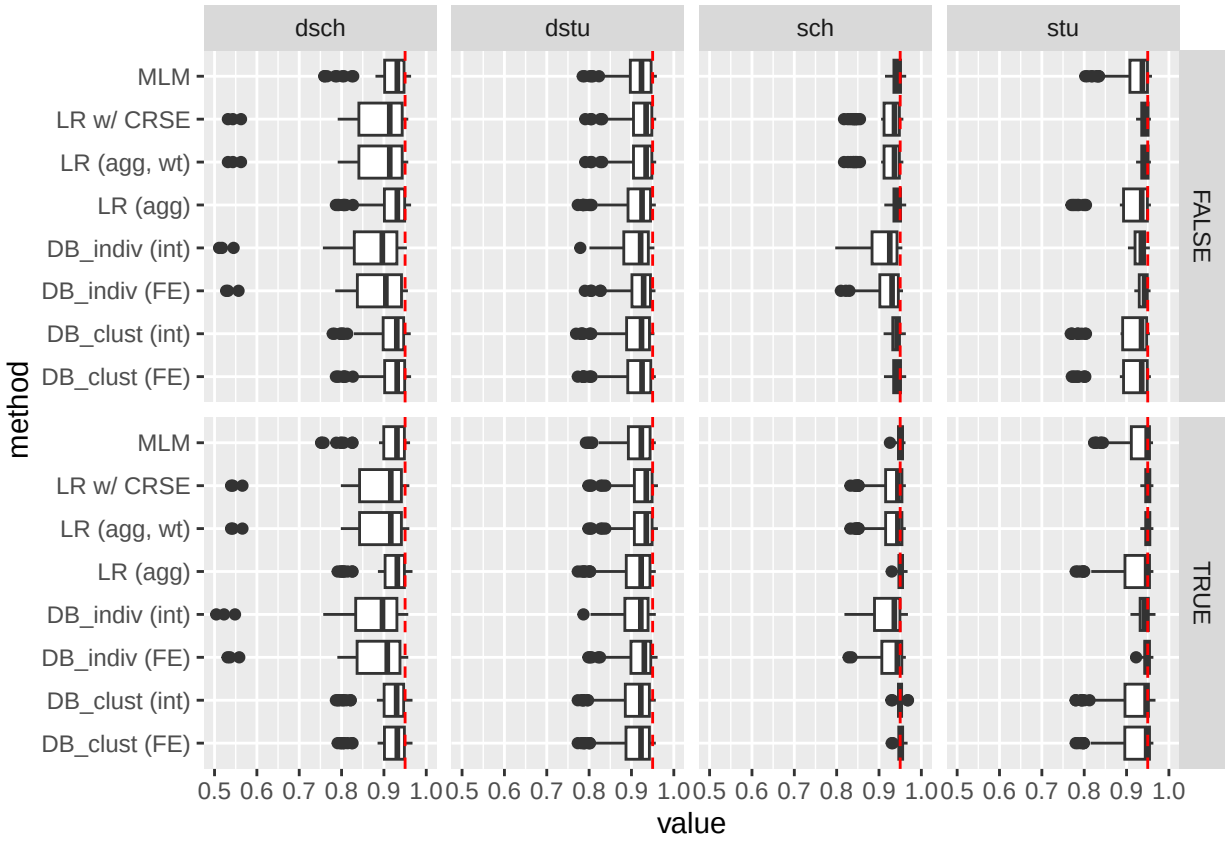
Coverage

Is generally sub-nominal.

```
sruns %>%
  group_by(method) %>%
  summarize(across(starts_with("cover"), mean)) %>%
  pivot_longer(starts_with("cover")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  ggplot(aes(x=method, y=value)) +
  geom_col() +
  facet_grid(finite ~ estimand) +
  geom_hline(aes(yintercept=0.95),
             lty="dashed", color="red") +
  coord_flip(ylim = c(0.5, 1))
```




```
sruns %>%
  pivot_longer(starts_with("cover")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  ggplot(aes(x=method, y=value)) +
  geom_boxplot() +
  facet_grid(finite ~ estimand) +
  geom_hline(aes(yintercept=0.95),
             lty="dashed", color="red") +
  coord_flip(ylim = c(0.5, 1))
```


dsch
dstu
sch
stu
FALSE
TRUE
method
MLM
LR w/ CRSE
LR (agg, wt)
LR (agg)
DB_indiv (int)
DB_indiv (FE)
DB_clust (int)
DB_clust (FE)
0.5 0.6 0.7 0.8 0.9 1.0
value

Next steps

A few initial impressions:

1. All of the true estimands tend to be super close to the overall average of 0.2.

```
sruns %>%
  summarize(across(starts_with("ATE."), mean))
```

```
##   ATE.super.district ATE.super.school ATE.super.student ATE.super.dist.student
## 1                0.2         0.216323         0.2327817              0.2159318
##   ATE.super.dist.school
## 1             0.2001122
```

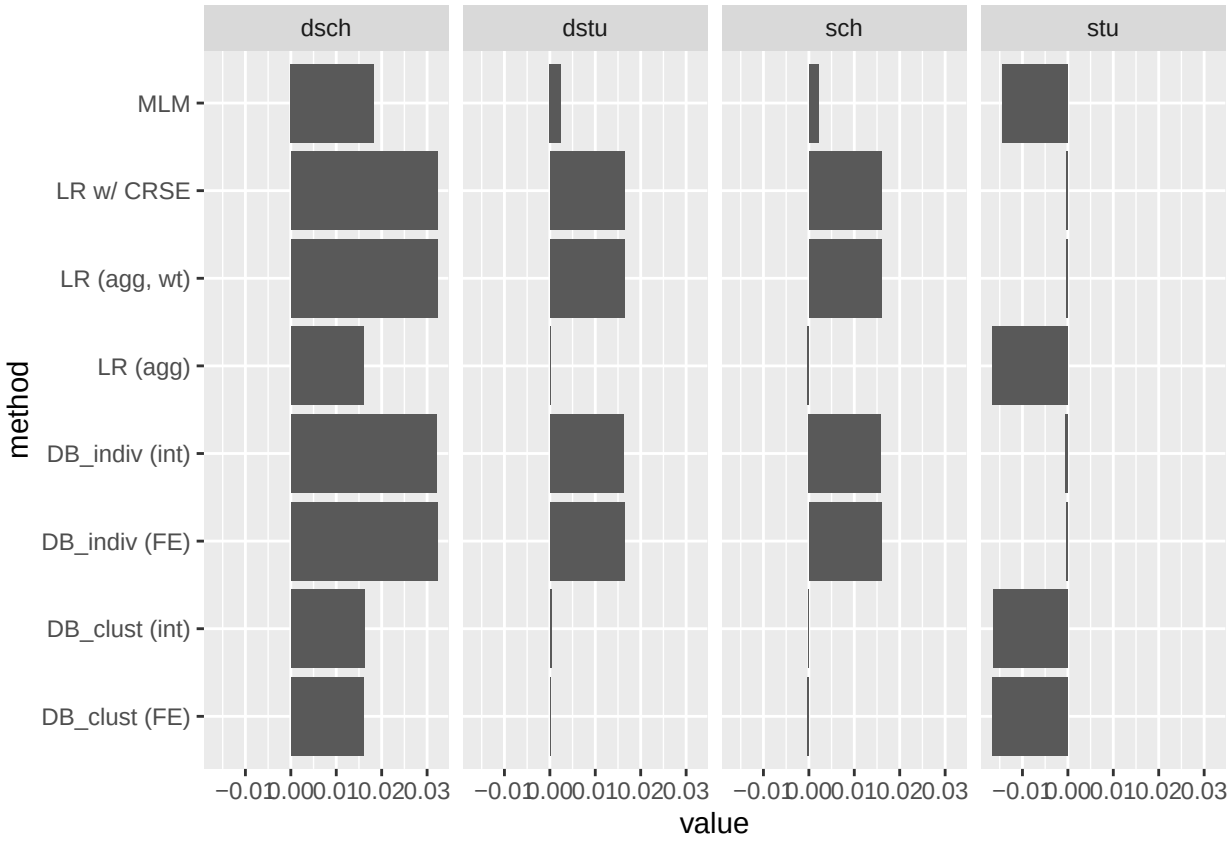
```
sruns %>%
  summarize(across(starts_with("ATE."), sd))
```

```
##   ATE.super.district ATE.super.school ATE.super.student ATE.super.dist.student
## 1                  0       0.02295606        0.03257283             0.02233689
##   ATE.super.dist.school
## 1           0.000367515
```

Presumably more torturing of the simulation parameters can lead to more extreme results, which I'll do next. This will also help clarify what the methods target.

2. Some strange bias results. While average *absolute* bias follows as we'd expect, average bias produces:

```
sruns %>%
  group_by(method) %>%
  summarize(across(starts_with("bias"), ~mean(.))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  filter(finite) %>%
  ggplot(aes(x=method, y=value)) +
  geom_col() +
  facet_grid( ~ estimand) +
  coord_flip()
```



Here, the district-school-weighted, school-weighted, and student-weighted results remain the same, but the district-student-weighted results strangely flip such that the school-weighted estimators appear to perform better for the true district-student-weighted estimand. I'll need to think a bit about what's going on here.

3. Other ideas?